Semantically-Augmented Graphic Libraries as Visual Standards for Anatomy and Phenotypes

Melissa Clarkson[1,2], Todd Detwiler[2], Noah Perry[2]

[1] Division of Biomedical Informatics [2] Institute for Biomedical Informatics University of Kentucky

We develop evidence-based graphic libraries that depict anatomy, phenotypes, and malformations. The graphics serve as visual standards that will clarify communication among researchers, clinicians, patients, curators, and others. They are also designed for use as visual assets in intelligent information systems. The graphics are augmented with semantics that describe the graphic itself, identify similar graphics, and link the graphics to external terminologies and ontologies.

endless FORMS studio

https://endlessforms.info

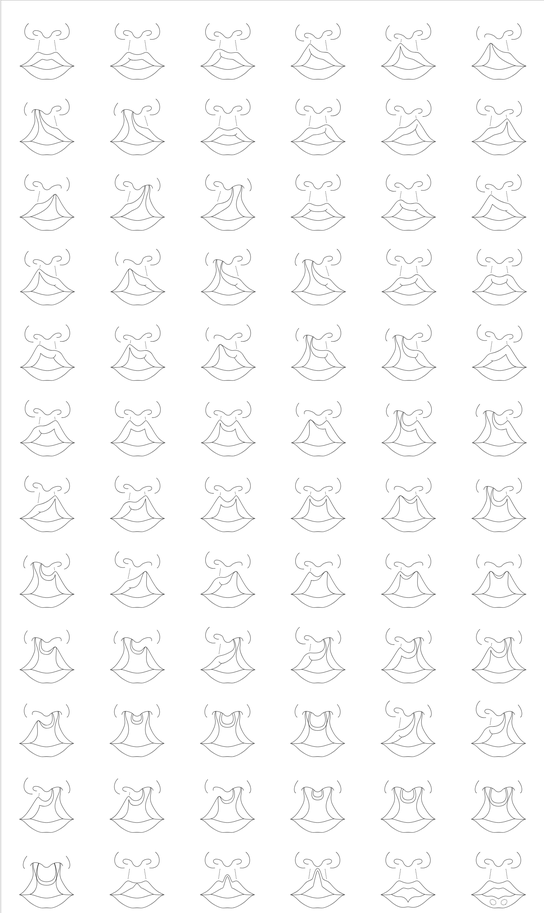
MD Clarkson, S Roggenkamp, LT Detwiler. 2025. Developing libraries of semantically-augmented graphics as visual standards for biomedical information systems. Journal of Biomedical Informatics. 163:104804.

Graphic libraries

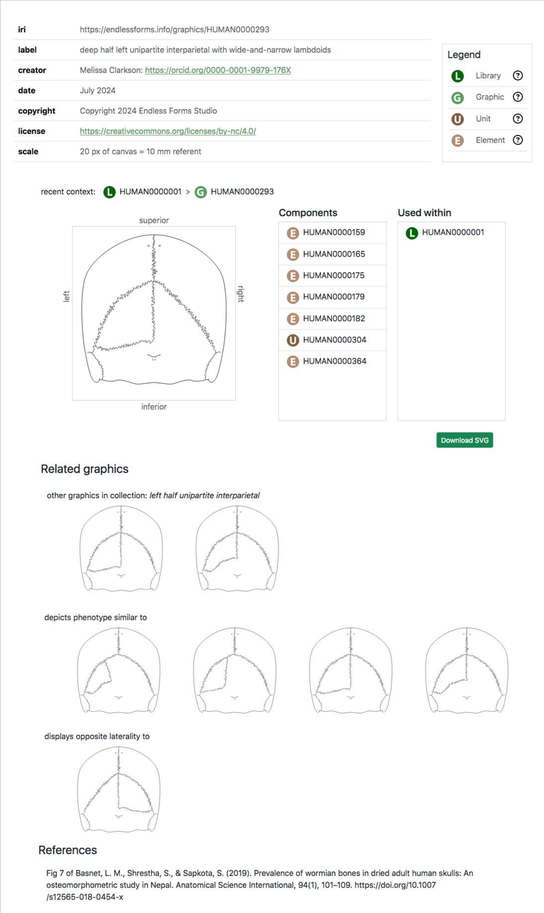
Scalable vector graphics (SVGs), each with an identifier in the form of an IRI that serves as a URL to an information page

https://graphics.endlessforms.info

The library of cleft lip graphics



Information page for one of the interparietal variant graphics



Graphic libraries in progress

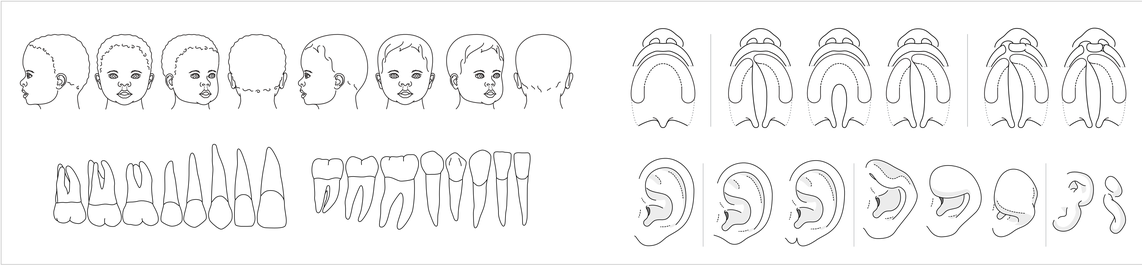
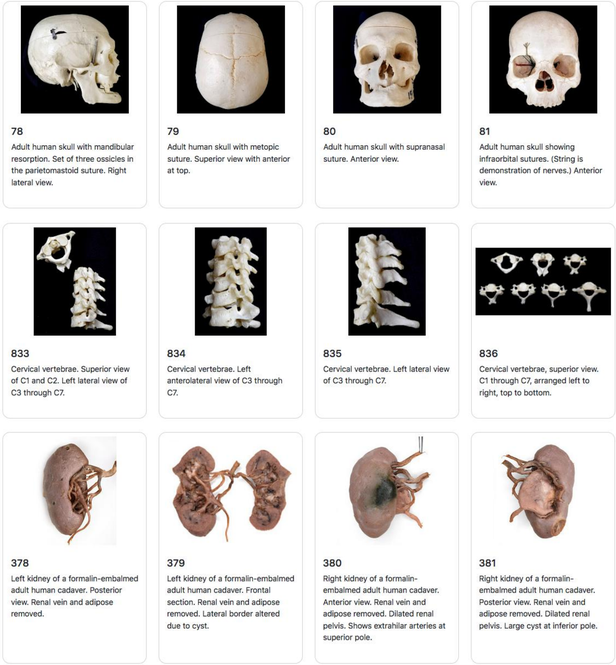


Image repository

Photos of specimens that serve as evidence for graphics

https://images.endlessforms.info



Graphic Descriptor Ontology (GDO)

Provides semantics for the graphics, including the anatomical view depicted, the representational purpose of each line, and related graphics

https://gdo.endlessforms.info

Phenotype browser

Provides links to external terminologies and ontologies, including ICD-11, Human Phenotype Ontology, Disease Ontology, MeSH

https://pheno.endlessforms.info

Public API

Enables retrieval of graphics and metadata about a graphic

https://graphics.endlessforms.info/api-info